NUR E JANNAT ASHA

Software Engineer | C++, Python & High-Performance Systems

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Work Authorization: Full Finnish Work Rights (LUT M.Sc. Graduate) | Immediate Availability | Aalto University F.E.C. Eligible

PROFESSIONAL SUMMARY

Software Engineer with a Master of Science in Software Engineering (**4.24/5.00 CGPA, Distinction**) from LUT University, Finland. Proven engineering track record in modern C++ (C++17), Python, Linux environments, and distributed backend architectures. Demonstrated success architecting deterministic finite state machines, accelerating high-performance numerical workloads with CUDA GPU parallelization (achieving 60% runtime reductions), and delivering production-ready software backed by automated CI/CD and rigorous unit testing. Accepted into the Aalto University F.E.C. (Further Educated with Companies) recruitment training program; available immediately for direct hire or subsidized company placement.

TECHNICAL SKILLS

Programming Languages: C++ (C++17/20, STL, RAII, Smart Pointers), C, Python (3.10+, FastAPI, NumPy, Pandas), SQL (PostgreSQL, SQLite), Bash, JavaScript (React 18), CUDA

Architecture & Systems: Object-Oriented Design (SOLID), Finite State Machines (FSM), Concurrency & Multithreading, Memory Optimization, RESTful API Design, Microservices

DevOps, Cloud & Tooling: Linux/Unix (POSIX), Docker (multi-stage builds), GitHub Actions CI/CD, CMake, Git, GDB, Valgrind, Postman

Testing & Quality: GoogleTest, PyTest, Test-Driven Development (TDD), Static Analysis (Cppcheck, Flake8), Code Review, Agile / Scrum

Domain Competencies: 5G Network Slicing (3GPP QoS/URLLC), LLM Verification & RAG Pipelines, Embedded Sensor Data Acquisition, Machine Control Logic

Languages: English (Fluent / Full Professional Working Proficiency), Finnish (A1 - active weekly coursework), Bengali (Native)

TECHNICAL PROJECTS

Deterministic Traffic Light Controller (C++17, CMake, GoogleTest) **github.com/Jannat332/traffic-light-controller**

Safety-Critical Embedded Finite State Machine Architecture

- Engineered a deterministic, cycle-accurate 4-phase intersection traffic controller in modern C++17 enforcing strict safety invariants (mathematically preventing conflicting green lights under all conditions).
- Implemented asynchronous emergency vehicle preemption and pedestrian walk timers with thread-safe synchronization and zero race conditions.
- Built an automated test suite achieving 100% test pass rate across edge state transitions, emergency overrides, and yellow-clearance timings.
- Structured with modular CMake configuration, RAII memory management, and zero raw pointers for high reliability.

LLM Parallel Reasoning & Verification Engine (Python 3.11, RAG, Docker) **github.com/Jannat332/llm-parallel-reasoning**

Dual-Stream Semantic Retrieval & Dynamic Contradiction Scoring

- Architected a dual-stream reasoning pipeline combining dense vector retrieval (RAG) with relational knowledge-graph validation, reducing factual hallucination rates by 42% on multi-hop benchmarks.
- Designed a dynamic confidence-scoring fusion algorithm that detects and flags factual contradictions in model outputs prior to client delivery.
- Containerized the entire evaluation workflow using Docker, enabling reproducible benchmark execution with single-command deployment.

5G Network Slice QoS Verification Demo (Python, 3GPP Specs) **github.com/Jannat332/5g-network-slice-demo**

Telecom SLA Simulation: URLLC Mission-Critical vs. eMBB High-Throughput

- Built a Python-based 3GPP-aligned network slicing simulator modeling dynamic resource allocation between ultra-low-latency URLLC and high-bandwidth eMBB traffic.
- Implemented strict priority queue scheduling ensuring sub-1ms packet latency for mission-critical industrial slices under heavy packet congestion.
- Engineered automated SLA verification with immediate anomaly alerts when packet loss or jitter thresholds are violated.

Production CI/CD Matrix Pipeline Template (GitHub Actions, Docker) **github.com/Jannat332/github-actions-ci-template**

Automated Multi-Platform Quality Assurance & Release Automation

- Constructed a reusable GitHub Actions CI/CD matrix pipeline executing cross-platform builds, automated linting, security vulnerability scans, and unit tests.
- Authored multi-stage Dockerfiles achieving a 65% image size reduction via Alpine runtime base layers, non-root security boundaries, and layer caching.

PROFESSIONAL EXPERIENCE

Junior Software Engineer | Spectrum IT Solutions **2021 – 2022**

Linux-Based Software Development & Backend Services

- Engineered, refactored, and maintained production software components and data endpoints using C, C++, and Python in Linux environments.
- Conducted root-cause defect analysis and memory optimization using GDB and Valgrind, eliminating memory leaks and race conditions in mission-critical modules.
- Collaborated within an Agile / Scrum framework; authored automated unit tests, actively participated in peer code reviews, and managed version control through Git branches and PR workflows.
- Optimized backend database queries and caching layers, improving data transaction response times by 25%.

Research Assistant (Computational Systems) | North South University **2023 – 2024**

High-Performance Computing, Numerical Modeling & GPU Acceleration

- Parallelized computational physics models using C++, Python, and CUDA GPU kernels, achieving up to 60% runtime acceleration compared to serial baselines.
- Automated large-scale numerical data extraction, convergence validation, and statistical analysis pipelines.
- Co-authored 3 research papers published in peer-reviewed Elsevier scientific journals based on the computational findings.

EDUCATION & QUALIFICATIONS

Master of Science in Software Engineering and Digital Transformation **2024 – 2025**

LUT University, Lappeenranta, Finland • ***CGPA: 4.24 / 5.00 (Distinction, Top 5%)***

- Curriculum: Advanced Software Architecture, Distributed Systems, Cloud Computing, Quality Assurance, Agile Engineering.

Bachelor of Science in Computer Science and Engineering **2016 – 2021**

North South University • ***CGPA: 3.44 / 4.00***

- Focus on Algorithms, Data Structures, Operating Systems, C/C++ Systems Programming. Bachelor thesis published in IEEE Xplore.

Aalto University F.E.C. (Further Educated with Companies) Program **Intake 2025 – 2026**

Aalto University, Espoo, Finland

- Accepted participant. Subsidized 6-month recruitment traineeship program transitioning into a permanent software engineering contract.

SELECTED SCIENTIFIC PUBLICATIONS

- **IEEE ICOEI 2021:** "Design of Low-Cost Sensor Embedded System for Real-Time Physiological Signal Processing." DOI: 10.1109/ICOEI51242.2021.9452873
- **Elsevier International Journal of Thermofluids (2024):** Numerical modeling and computational analysis of thermal transport dynamics. DOI: 10.1016/j.ijft.2024.100650
- **Elsevier Heliyon (2024):** Computational modeling of fluid-structure thermal phenomena. DOI: 10.1016/j.heliyon.2024.e32306
- **Elsevier Case Studies in Thermal Engineering (2024):** Finite element simulation and parametric optimization. DOI: 10.1016/j.csite.2024.104273